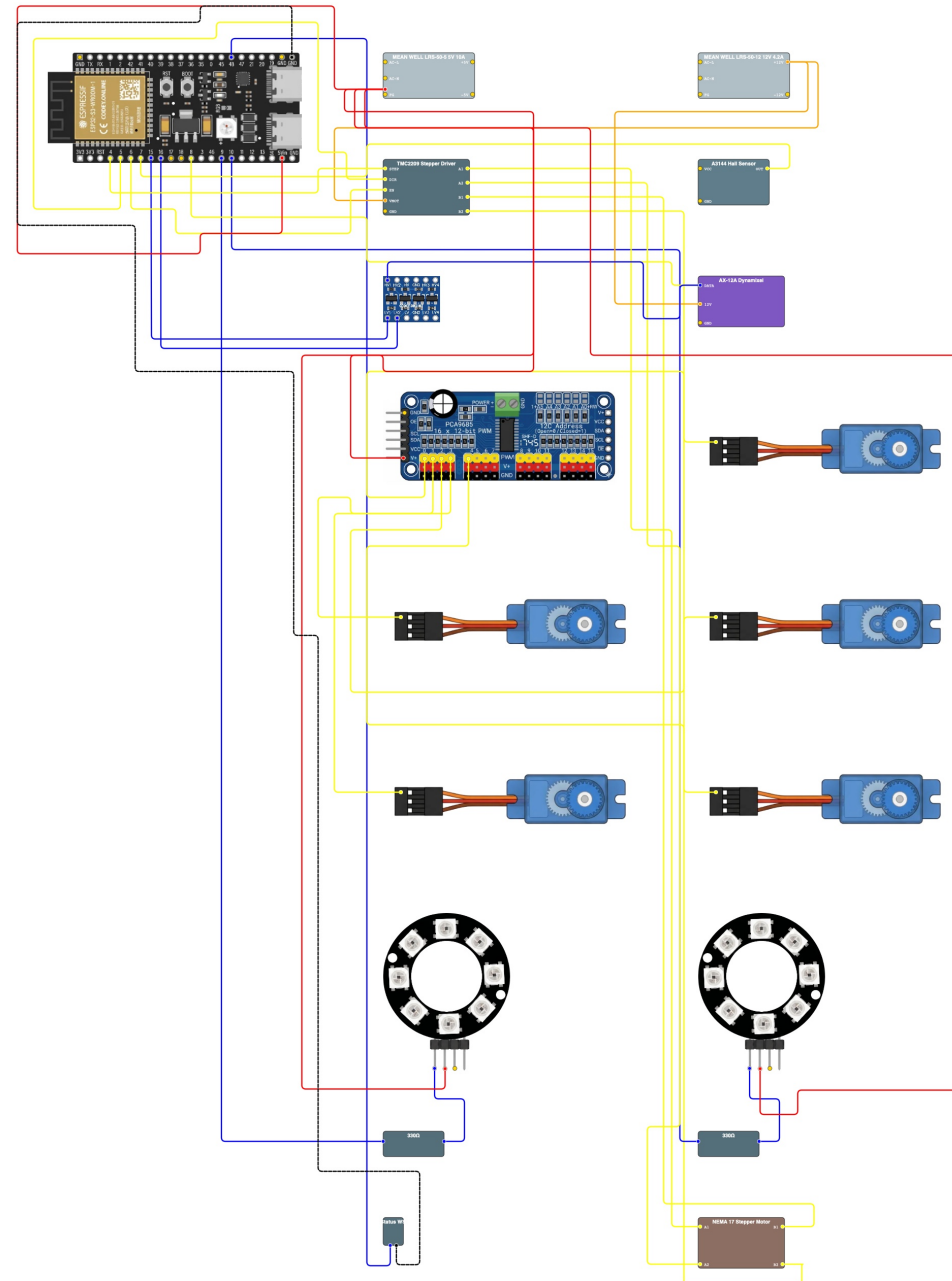




Wiring connections

WIRE	FROM	TO
	MEAN WELL LRS-50-5 5V 10A +5V	ESP32-S3 N16R8 DevKitC 5Vin
	MEAN WELL LRS-50-5 5V 10A +5V	Pololu Mini Maestro 24ch V+
	MEAN WELL LRS-50-5 5V 10A +5V	WS2812 Rear LED Ring 16 LEDs 5V
	MEAN WELL LRS-50-5 5V 10A +5V	WS2812 Front LED Ring 35 LEDs 5V
	MEAN WELL LRS-50-5 5V 10A GND	ESP32-S3 N16R8 DevKitC GND
	MEAN WELL LRS-50-5 5V 10A GND	Pololu Mini Maestro 24ch GND
	MEAN WELL LRS-50-5 5V 10A GND	WS2812 Rear LED Ring 16 LEDs GND
	MEAN WELL LRS-50-5 5V 10A GND	WS2812 Front LED Ring 35 LEDs GND
	MEAN WELL LRS-50-12 12V 4.2A +12V	TMC2209 Stepper Driver VMOT
	MEAN WELL LRS-50-12 12V 4.2A +12V	AX-12A Dynamixel 12V
	MEAN WELL LRS-50-12 12V 4.2A GND	TMC2209 Stepper Driver GND
	MEAN WELL LRS-50-12 12V 4.2A GND	AX-12A Dynamixel GND
	MEAN WELL LRS-50-12 12V 4.2A GND	ESP32-S3 N16R8 DevKitC GND.1
	MEAN WELL LRS-50-12 12V 4.2A GND	A3144 Hall Sensor GND
	ESP32-S3 N16R8 DevKitC 4	TMC2209 Stepper Driver STEP
	ESP32-S3 N16R8 DevKitC 5	TMC2209 Stepper Driver DIR
	ESP32-S3 N16R8 DevKitC 6	TMC2209 Stepper Driver EN
	ESP32-S3 N16R8 DevKitC 7	A3144 Hall Sensor OUT
	ESP32-S3 N16R8 DevKitC 8	AX-12A Dynamixel DATA
	ESP32-S3 N16R8 DevKitC 9	3 3 0 © 1
	3 3 0 © 2	WS2812 Rear LED Ring 16 LEDs DI
	ESP32-S3 N16R8 DevKitC 10	3 3 0 © 1
	3 3 0 © 2	WS2812 Front LED Ring 35 LEDs DI
	ESP32-S3 N16R8 DevKitC 15	Tri-state buffer / level shifter LV1
	ESP32-S3 N16R8 DevKitC 16	Tri-state buffer / level shifter LV2
	Tri-state buffer / level shifter HV1	AX-12A Dynamixel DATA
	ESP32-S3 N16R8 DevKitC 17	Pololu Mini Maestro 24ch RX
	ESP32-S3 N16R8 DevKitC 18	Pololu Mini Maestro 24ch TX
	ESP32-S3 N16R8 DevKitC 48	Onboard Status WS2812 LED ANODE +
	Onboard Status WS2812 LED CATHODE -	ESP32-S3 N16R8 DevKitC GND.2
	Pololu Mini Maestro 24ch CH0	MG996R Servo Lower Arm PWM
	Pololu Mini Maestro 24ch CH1	MG996R Servo Elbow PWM
	Pololu Mini Maestro 24ch CH2	MG90S Servo Neck Pan PWM
	Pololu Mini Maestro 24ch CH3	MG996R Servo Spare 1 PWM
	Pololu Mini Maestro 24ch CH4	MG996R Servo Spare 2 PWM
	TMC2209 Stepper Driver A1	NEMA 17 Stepper Motor A1
	TMC2209 Stepper Driver A2	NEMA 17 Stepper Motor A2
	TMC2209 Stepper Driver B1	NEMA 17 Stepper Motor B1
	TMC2209 Stepper Driver B2	NEMA 17 Stepper Motor B2